RAYHAN MUHAMMAD ALFARIZI

Software Developer | Backend Developer | Machine Learning & IoT Enthusiast

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Electrical Engineering graduate from Diponegoro University with hands-on experience in full-stack web development, backend development, IoT, and machine learning through internships and academic projects. Proficient in developing RESTful APIs, backend services, SQL/NoSQL databases, Docker, Git, and modern web technologies. Passionate about building scalable software and continuously learning cloud-native technologies.

EDUCATION

Diponegoro University **2022 – 2026**

- Bachelor of Electrical Engineering | GPA: 3.87 / 4.00
- Relevant Coursework: Algorithms and Programming, Data Structures, Object-Oriented Programming, Software Engineering, Database, Operating Systems, Web Development, Modern Programming Methods

TECHNICAL SKILLS

Programming Languages
- Python, PHP, JavaScript, TypeScript, C/C++, SQL

Backend
- FastAPI, Laravel, CodeIgniter 4, REST API

Frontend
- Next.js, React, Livewire, Tailwind CSS

Database
- PostgreSQL, MySQL/MariaDB, Firebase, Supabase, InfluxDB

Machine Learning
- TensorFlow, Keras, Autoencoder, Anomaly Detection

IoT & Messaging
- ESP32, MQTT

Cloud & DevOps
- Docker, DigitalOcean (VPS deployment), Git, GitHub, Linux

EXPERIENCE

Web Development Intern **Jan – Feb 2025**
PT Des Teknologi Informasi
- Developed backend features for a Guest Management System using CodeIgniter 4.
- Built RESTful APIs for guest registration and scheduling.
- Implemented authentication and role-based authorization.
- Collaborated with a two-person development team using Git and GitHub.
- Contributed to dashboard development and scheduling functionality.

Teaching Assistant – Algorithms and Programming **Feb – Jun 2025**
Diponegoro University
- Assisted laboratory sessions for 30 undergraduate students.
- Taught Python and C++ programming fundamentals.
- Guided students in debugging and problem-solving techniques.

Teaching Assistant – Electrical Circuits **Aug – Dec 2025**
Diponegoro University

- Assisted laboratory sessions for 20 undergraduate students
- Guided students in circuit experiments and measurement instruments.
- Evaluated and provided technical feedback on laboratory reports for 10 assigned students

PROJECTS

Autoencoder-Based Fault Detection & Diagnosis for Smart Precision Irrigation **Jan – Jul 2026**
Tech Stack: Python, FastAPI, Laravel, Livewire, MQTT, ESP32, Docker, MariaDB, InfluxDB

- Designed and implemented an end-to-end IoT monitoring and fault diagnosis platform.
- Developed backend services for real-time data collection and anomaly detection.
- Built REST APIs and integrated MQTT communication.
- Applied an Autoencoder model achieving 97.50% accuracy and 97.56% F1-score.
- Deployed containerized backend services on a DigitalOcean VPS using Docker.

Artikula: Reading Space **Feb 2026**
Tech Stack: Laravel, Livewire, PostgreSQL (Supabase)

- Developed a full-stack article publishing platform.
- Implemented authentication, search, filtering, analytics dashboard, and feedback features.
- Designed relational database using PostgreSQL.

Letsgobois Shop – E-Commerce Platform **Sep – Dec 2025**
Tech Stack: Next.js, Firebase, JavaScript

- Developed a full-stack e-commerce application.
- Implemented Google Authentication and Midtrans payment gateway.
- Developed role-based access management and responsive user interfaces.

LEADERSHIP

Chief Executive – Elektro Forge (E-Forge) **Aug – Sep 2024**
Electrical Engineering Student Association (HME), Diponegoro University

- Led a 5-member organizing committee to deliver a PCB Design Training for 54 participants.
- Coordinated planning, logistics, and stakeholder communication to ensure successful event execution.

CERTIFICATIONS

- Python Essentials 2 – Cisco Networking Academy
- Python Essentials 1 – Cisco Networking Academy
- C Essentials 1 – Cisco Networking Academy
- Machine Learning for Beginners — Dicoding Indonesia
- Data Analysis with Python — Dicoding Indonesia
- Basic SQL — Dicoding Indonesia

LANGUAGES

- Indonesian – Native
- English – Professional Working Proficiency
  SEU TOEFL Prediction Test (Equivalent Score: 513)